# Luvic Arenosols:

These are Sandy soils with a **luvic horizon**: a subsurface clay-enriched horizon (Bt horizon) formed by downward movement (illuviation) of clay. Unlike other Arenosols, they combine sandy surface layers with a clay-rich subsoil.

**Drainage:**

- The Clay-enriched sub-surface may impede drainage and cause temporary waterlogging in wet seasons.

**Depth:**

- Arenosols are usually deep, often deeper than 100 cm,
- The Effective rooting depth may be reduced if the clay horizon is dense/compacted.

**Workability:**

- Because of the sandy texture, Arenosols are **easy to work** mechanically: they are light, not very cohesive, don't require much force to till etc. This is good for tillage.
- ***Con:*** they're prone to wind erosion, shifting, collapse of structure under rain, etc. Also seedling establishment can be difficult in very loose sandy soils.
- The Subsoil clay may be harder, sticky when wet, and compact when dry making deeper cultivation difficult.

**Nutrient Richness & Cation Exchange Capacity (CEC):**

- Luvic arenosols have a higher nutrient retention than other Arenosols due to clay accumulation.
- Still often low in organic matter, nitrogen, and phosphorus.
- Base saturation may be moderate if the clay has active exchange sites.

**pH:**

- slightly acidic to neutral.
- Depends on clay type and degree of leaching.

**Management:**

- Build up **organic matter** in topsoil to improve fertility and water-holding.
- Break up compacted clay layers (deep tillage, subsoiling) where possible.
- Use balanced fertilizers: N and P especially.
- Maintain crop residues or cover crops to reduce erosion and sustain fertility.
- Favor crops tolerant to variable moisture (maize, millet, groundnuts).